Section 2.1 — Monomials and Polynomials

Math 103: College Algebra

Exponents

For each expression below:

a. Identify the base and the exponent.

b. Rewrite as repeated multiplication.

c. Evaluate by multiplying, then check your answer using the exponent key on your calculator.

1. 2^3
2. 3^4
3. $(-4)^2$
4. 5^3
5. $(-1)^5$

For each of the expressions below:

a. Rewrite using a positive exponent.

b. Evaluate using your rewritten expression, then check your answer using your calculator. Answer as a fraction.

6. 3^{-2}
7. 5^{-1}
8. $\left(\frac{2}{3}\right)^{-3}$
9. $(-4)^{-3}$
10. $\left(-\frac{1}{3}\right)^4$

Simplifying Monomials

Simplify each monomial below, applying exponent rules as appropriate.

11. b^2b^3
12. $2y^2 \cdot 5y^3z$
13. $\left(3x^2\right)^3$
14. $\left(-\frac{3}{2}x\right)^4$
15. $\left(2x^3\right)^4\left(2xy\right)^2$
16. $\frac{3}{8}x^2y \cdot \frac{4}{9}xy^2$
17. $6hk \cdot 7j^5k \cdot (-2)h^2y$

Using Polynomial Vocabulary

For each expression, decide whether it is

A. not a polynomial
B. a monomial
C. a binomial
D. a trinomial
E. a polynomial with 4 or more terms

18. $b + 2$
19. $-\dfrac{2}{3}tu$
20. x^3
21. $x^2 - 7xy - y^2$
22. 13
23. $-0.6xy^2$
24. $\dfrac{4m^2}{n}$
25. $4m - \dfrac{3}{n^2}$
26. $\frac{5}{8}ab$
27. $-0.9n^2p^5$
28. $7r^3 - 9r^2 + r - 6$
29. $\dfrac{5}{x^2}$
30. $5.7x^4y$
31. y^4
32. $x^2 + y^{-3}$
33. $2x + 1$

For each polynomial below:

a. Identify the leading term and leading coefficient, if any.

b. Give the degree of the polynomial.

c. Identify any that are constant, linear, quadratic, or cubic.

34. $6x + 8y + 3z$
35. $17.3x^2 - 3x + 2.1$
36. $18xy^3$
37. $x^3 - 4x^2 + 4x + 4$
38. $7k + 5$
39. 36
40. $25 - x^2$
41. $2u^2 + 5u^3 - u^7$

Give an example of each of the following.

42. A quadratic trinomial in one variable.
43. A homogenous cubic polynomial with four terms. (Use x and y as your variables.)
44. A linear binomial in one variable.
45. A cubic binomial in one variable.

Polynomial Functions

Let $f(n) = n^2 - 8n - 3$. Find:

46. $f(2)$
47. $f(-3)$
48. $f(\frac{1}{2})$
49. $f(0)$

Let $g(j) = 22j^3 + 5j^2 - 16j^4 + 21 - 14j$. Find:

50. $g(1)$
51. $g(-2)$
52. $g(\frac{1}{4})$
53. $g(-1)$